

# Assignment: Git Branch, PR and Local Merge

## Objective

Practice this complete flow:

```
create folder -> create file -> init -> commit -> connect remote -> push main -> create branch -> push branch -> raise PR -> merge PR -> pull main -> create another branch -> local merge -> push main
```

## Part 1: Create Folder and File

Right click on desktop and select open in terminal

Run:

```
mkdir git-branch-pr-practice
cd git-branch-pr-practice
code .
```

Create a file named:

```
notes.txt
```

Add this content:

```
This is my Git branch and PR practice file.
This file was created on the main branch.
```

Save the file.

## Part 2: Initialize Git and First Commit

Run:

```
git init
git status
git add .
git branch -M main
git commit -m "initial commit"
```

## Part 3: Connect Remote and Push Main

Create an empty GitHub repository.

Then copy your GitHub repository URL and run:

```
git remote add origin PASTE_GITHUB_URL_HERE
git push -u origin main
```

Check GitHub. The `notes.txt` file should be visible on the `main` branch.

## Part 4: Create a Branch Locally

Create and switch to a new branch:

```
git switch -c text-update
```

Check branch:

```
git branch
```

## Part 5: Modify, Stage, Commit

Open `notes.txt` and add:

This line was added from the text-update branch.

Run:

```
git status
git add .
git commit -m "updated notes from text-update branch"
```

## Part 6: Push Branch to GitHub

Run:

```
git push -u origin text-update
```

## Part 7: Raise PR and Merge on GitHub

Go to GitHub and create a Pull Request.

Use:

```
base: main
compare: text-update
```

Then click:

```
Create Pull Request
Merge Pull Request
Confirm Merge
```

## Part 8: Pull Latest Main to Local

Come back to terminal and run:

```
git switch main  
git pull origin main
```

Open `notes.txt` and confirm the `text-update` line is available locally.

## Part 9: Create Another Local Branch

Create a new branch:

```
git switch -c local-update
```

Open `notes.txt` and add:

```
This line was added from the local-update branch.
```

Run:

```
git status  
git add .  
git commit -m "added local update"
```

## Part 10: Merge Local Branch into Main

Switch to main:

```
git switch main
```

Merge the branch locally:

```
git merge local-update
```

Check `notes.txt`. The `local-update` line should be visible on `main`.

# Part 11: Push Updated Main to GitHub

Run:

```
git push origin main
```

Refresh GitHub and confirm all updates are visible in `notes.txt`.

## Full Command Flow

```
mkdir git-branch-pr-practice
cd git-branch-pr-practice
code .

# create notes.txt

git init
git status
git add .
git commit -m "initial commit"

git remote add origin PASTE_GITHUB_URL_HERE
git branch -M main
git push -u origin main

git switch -c text-update
# modify notes.txt

git status
git add .
git commit -m "updated notes from text-update branch"
git push -u origin text-update

# raise PR and merge it on GitHub
```

```
git switch main
git pull origin main

git switch -c local-update
# modify notes.txt

git status
git add .
git commit -m "added local update"

git switch main
git merge local-update
git push origin main
```

## Final Check

Verify:

- `main` branch pushed to GitHub.
- `text-update` branch pushed to GitHub.
- Pull Request created and merged.
- Latest `main` pulled locally.
- `local-update` branch created locally.
- `local-update` merged into `main` locally.
- Updated `main` pushed to GitHub.

Final memory lines:

```
PR merge happens on GitHub.
Local merge happens in our system.
After PR merge, pull main locally.
After local merge, push main to GitHub.
```